

MiniProject_Report_Varga

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Introduciton

Before I began my project proposal, while cruising through the suggested datasets for this project, I found the UFO sightings dataset. This was immediately interesting so I looked further into it. The dataset is massive, covering the entire world, with a total of over 80,000 observations over more than a hundred years. I submitted a rough proposal and waited to continue working on the project. After about two weeks, I was contacted by a fellow student Christine Lazina and we decided to team up on the project.

I wanted to try and use most, if not all, of the analyses methods and tools that we have learned throughout the term. I believed using a simple point dataset would allow the vast majority of tools. Though Christine and I would later discover that the UFO dataset didn't fit the criteria for many of the analysis methods.

For the project itself, we decided to go with a data-driven approach, where we began looking at and working with the data, while forming questions that we could answer. Due to the data being very extensive, we decided to limit our scope to California. We also chose California due to it having the highest number of observations. Additionally, I believe California to have a significant military presence, possibly implying that there could be a correlation between military activity and UFO sightings.

Project Questions

Is there a correlation between military bases/airports and UFO sightings?

Do seasons have an effect on the number of UFO sightings?

Methods

The UFO sightings dataset is from the National UFO Reporting Center (2014). The dataset has 11 variables. Most are categorical, but others are numeric. The data covers all the important aspects of a sighting, such as the precise location (lat long) as well as general location, shape, length of observation, and date and time of. Before even beginning the analysis, Christine and I agreed that we would need more spatial data to use in conjunction with the UFO sightings. We first added a state border with the spData package. We then added a California cities, airports, and military bases shapefile. These external shapefiles were easily found for free online via the California State Geoportal.

First, we must load the necessary libraries.

```
library(dplyr)      # working with data frames and especially the filter() function
library(sf)         # working with sf objects
library(spData)     # state borders
library(tmap)       # the mapping tool of choice
```

```
library(spatstat) # spatial point pattern analysis
library(tidyverse) # month() function
```

The UFO sightings data originated as a .csv spreadsheet. To import it, we simply use the read.csv function, which imports it as a data frame object (we will later convert this into an sf). We also read-in the three shapefiles: cities, airports, and military bases.

```
# UFO dataset
ufo <- read.csv("./UFOscrubbed.csv")

# California cities shapefile
cali_cities <- read_sf("./CA_Cities_SHP/CalifCities.shp")

# California airports shapefile
cali_airports <- read_sf("./Public_Airport/AVN_Public_Airports.shp")

# Military Bases shapefile
bases <- read_sf('./Military_Bases/military_bases_ds_military_bases_2023_09_20.shp')
```

Since the UFO sightings dataset is much too big, we filter it to only look at the state of California. This is done simply with the filter() function. Since some entries don't have recorded states, or a country, I added an & clause to the function to filter for both state and country. This ensured no observations with missing state or country fields. Filtering the CA state border from spData was the same process, as well as filtering the cities shapefile to only major cities. Finally, we filtered the military bases to only look at bases inside California.

```
# Filter UFO dataset for California only
ufo_ca <- filter(ufo, country == 'us' & state == 'ca')

# Filter CA from us_states
cali <- filter(us_states, NAME == 'California')

# Filter cali_cities to only major cities
cali_mcities <- filter(cali_cities, POP_CL > 8)

# Filter military bases dataset for California only
bases_ca <- filter(bases, statenamec == 'CA')
```

An important step in answering the second research question is assigning observations to their respective seasons. To do this, we covered the 'datetime' variable to a M-D-Y format, which will make it easier to filter into seasons. The second block of code simply converts the UFO dataframe into an sf object, setting the coordinates to the latitude and longitude variables within the dataframe.

```
# Convert data in UFO datetime column to Date format
ufo_ca <- ufo_ca %>%
  mutate(Date = as.Date(ufo_ca$datetime, "%m/%d/%Y"), .after = datetime)

# Convert ufo_ca dataframe into a sf object
ufo_ca_sf <- st_as_sf(ufo_ca, coords = c("longitude", "latitude"))
```

To minimize future headaches, we transformed all data to California Albers coordinate system, EPSG:3310. But first, we have to define the CRS before transforming the UFO sightings.

```
# Define ufo_ca_sf CRS as NAD83 (alternate: WGS84 is 4326)
ufo_ca_sf <- st_set_crs(ufo_ca_sf, 4269)
```

```
st_crs(ufo_ca_sf)
```

```
# Transform all dataset CRS's to projected California Albers CRS
ufo_ca_sf <- st_transform(ufo_ca_sf, crs=3310)
cali_mcities <- st_transform(cali_mcities, crs=3310)
cali_airports <- st_transform(cali_airports, crs=3310)
bases_ca <- st_transform(bases_ca, crs=3310)
```

Assigning Seasons to Observations

It took some trial and error to find an elegant way of using the 'datetime' variable to assign seasons to each observation. I knew I had to use the filter() one way or another, but my code in the beginning was quite messy. Fortunately, I got it to a point where it was as clean as I could get it. Using 'or' statements in the filter function to create a new object with observations from only the selected months. I used December, January, and February for winter, March, April, and May for spring, June, July, and August for summer, and September, October, and November for fall. To combine all four objects back into a dataframe, I simply used the rbind() function to vertically join them.

```
# split dates into four seasons in separate df
winter <- filter(ufo_ca_sf, (month(ufo_ca_sf$Date) == 12 | (month(ufo_ca_sf$Date) == 1 |
(month(ufo_ca_sf$Date) == 2))))
winter$season <- rep(c("winter"), times=2028)

spring <- filter(ufo_ca_sf, (month(ufo_ca_sf$Date) == 3 | (month(ufo_ca_sf$Date) == 4 |
(month(ufo_ca_sf$Date) == 5))))
spring$season <- rep(c("spring"), times=1913)

summer <- filter(ufo_ca_sf, (month(ufo_ca_sf$Date) == 6 | (month(ufo_ca_sf$Date) == 7 |
(month(ufo_ca_sf$Date) == 8))))
summer$season <- rep(c("summer"), times=2639)

fall <- filter(ufo_ca_sf, (month(ufo_ca_sf$Date) == 9 | (month(ufo_ca_sf$Date) == 10 |
(month(ufo_ca$Date) == 11))))
fall$season <- rep(c("fall"), times=2332)

# bind all four back together
ufo_ca_sf2 <- rbind(winter, spring, summer, fall)
```

Point Pattern Analysis

Unfortunately, due to severe difficulties with the data, we were unable to find ways to complete all the analyses that we had aimed for. However, point pattern analyses worked well, and provided us with useful figures and data. To begin, we must convert the sf object to a ppp object. In this code, we had to specify the geometry in the sf for the ppp object. If this wasn't done, the conversion would fail.

```
# Convert CA UFO Sightings sf object to ppp object
ufo_ca_ppp <- select(ufo_ca_sf, state:geometry) %>%
  as.ppp()
```

Kernal Density Estimation (KDE)

Beginning with a simple KDE analysis, which is a slightly more complex density-based analysis tool compared to something like quadrats. It works by computing density of points for an area with a moving window called a kernal. It can create a typical heat map-like figure that is easily understandable.

Christine determined a sigma value of about 15000 was optimal, but others are shown as well for comparison. For all KDEs, contour lines are added to help interpretation and readability.

```
# bandwidth 10,000
d10K <- density(ufo_ca_ppp, sigma = 10000)
plot(d10K, main="sigma=10000")
contour(d10K, add = T)

# bandwidth 15,000
d15K <- density(ufo_ca_ppp, sigma = 15000)
plot(d15K, main="sigma=15000")
contour(d15K, add = T)

# bandwidth 20,000
d20K <- density(ufo_ca_ppp, sigma = 20000)
plot(d20K, main="sigma=20000")
contour(d20K, add = T)
```

Here we use the default sigma, as well as functions to automatically determine an appropriate sigma value.

```
# use default sigma (not specified)
def <- density(ufo_ca_ppp)
plot(def, main="sigma=default")
contour(def, add=T)

# specify sigma with bw.diggle() function
d_diggle <- density(ufo_ca_ppp, bw.diggle(ufo_ca_ppp))
plot(d_diggle, main=paste("sigma=", as.character(round(bw.diggle(ufo_ca_ppp),2))))
contour(d_diggle, add=T)

# specify sigma with bw.ppl() function
d_ppl <- density(ufo_ca_ppp, bw.ppl(ufo_ca_ppp))
plot(d_ppl, main=paste("sigma=", as.character(round(bw.ppl(ufo_ca_ppp),2))))
contour(d_ppl, add=T)
```

Ripley's K/L Functions

Moving on to a distance-based method (KDE was density-based), Ripley's K and L functions are a simple way to analyze distances between points. They are showcased as a plot that is easily readable. First, these functions count the number of neighboring points within given distances for all features. Then it sums the values at each distance (called a band). Both the K and L functions are very similar, but the L function is easier to understand due to it removing the curve from the plot line.

The code for either is very similar, with the only difference being ran with 'Kest' or 'Lest'. These functions are ran on the ppp object from earlier.

```
# Create a K-function for the UFO point data with confidence envelope
k_ufo <- envelope(ufo_ca_ppp, Kest, correction='best')

# Create an L-function for the UFO point data with confidence envelope
l_ufo <- envelope(ufo_ca_ppp, Lest, correction='best')
```

Spatial Multivariate Clustering

Both Christine and I were both hoping to use spatial multivariate clustering to identify clusters of multiple variables. I created the 'seasons' variable with the hope of using it here. Unfortunately, I was not aware that it requires numeric-only data, at least the method from the lab assignment. I tried many alterations of code to get it to work and got no where. Christine found an awesome article online website that showcased a method allowing for categorical variables. While I was able to get the blog's example to work, it didn't work with our data, as our dataset had been formatted differently. Regardless of my difficulties, the code is below.

Here is the page on clustering mixed data types in R. This alternative method works the same way, but uses a different combination of functions (from different packages as well) to achieve an end result. I would have loved to have the time and ability to fully learn this method.

<https://dpmartin42.github.io/posts/r/cluster-mixed-types>

```
# subset the data for the variables of interest
ufo_sel <- select(ufo_ca_sf2, c(1,2,7,11))
head(ufo_sel)

# create an empty vector that will hold 10 within-cluster-sum-of-squares
wcss <- numeric(10)

# run kmeans 10 times producing results for 1 to 10 possible clusters
for (i in 1:10) {
  # Fit the model: km.out
  km.out <- kmeans(ufo_sel[, -1], centers = i)
  # Save the within cluster sum of squares
  wcss[i] <- km.out$tot.withinss
}

# Put the data in a tibble and plot a scree plot
wcss_df <- tibble(clusters = 1:10, wcss = wcss)

# have not even gotten here yet
scree_plot <- ggplot(wcss_df, aes(x = clusters, y = wcss, group = 1)) +
  geom_point(size = 4) +
  geom_line() +
  scale_x_continuous(breaks = c(2, 4, 6, 8, 10)) +
  xlab('Number of clusters')
```

ANN Index

The ANN index code is quite simple. But a different function had to be used to calculate the area when calculated expected distances - this was due to the nature our ppp object.

```
# Calculate the nearest neighbor distances, and mean distance for the ufo point data
nnd_ufo <- nndist(ufo_ca_ppp)
mnnd_ufo <- mean(nnd_ufo)

# Calculate expected distances between NN
emnnd_ufo <- 0.5/sqrt(npoints(ufo_ca_ppp)/spatstat.geom::area(ufo_ca_ppp))

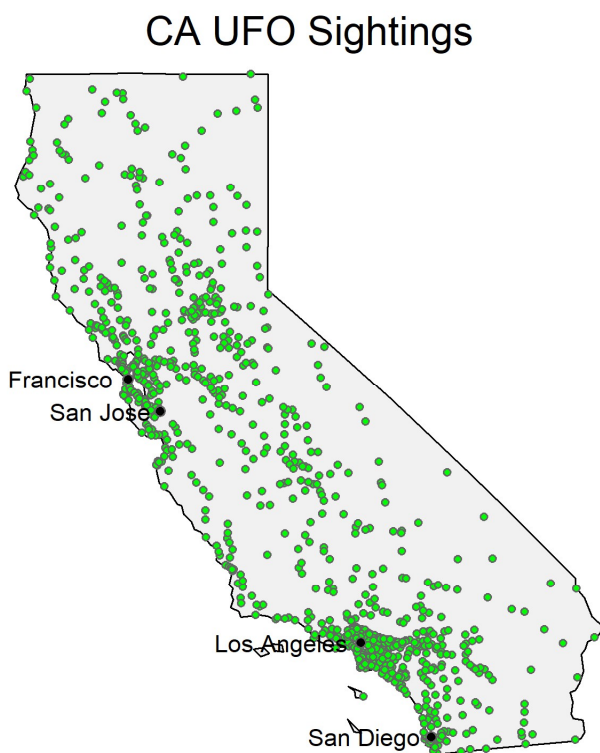
# Calculate the ANN ratio (ANN = 0.06721282)
ann_ufo <- mnnd_ufo/emnnd_ufo
```

Results

Even though we had significant troubles with several aspects and goals in this project, we still were able to create several valuable maps showcasing the information. Additionally, since all of the point pattern analyses worked properly, we have figures for that as well.

I'll begin by creating a basic map of all UFO sightings in California. For this project, Christine and I used the tmap package to create our maps. Personally, I find tmap to be the easiest package to work with for making maps. The code here is simple and doesn't require much explaining, but I will make a note of the tmap_options() function which has an argument to check sf shapes for validity. On another note, I don't know how to stop the text for the cities to not be cut off. The par(mar) argument doesn't seem to do anything, no matter what I set it to.

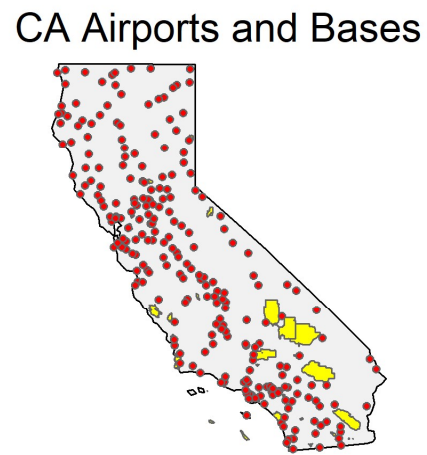
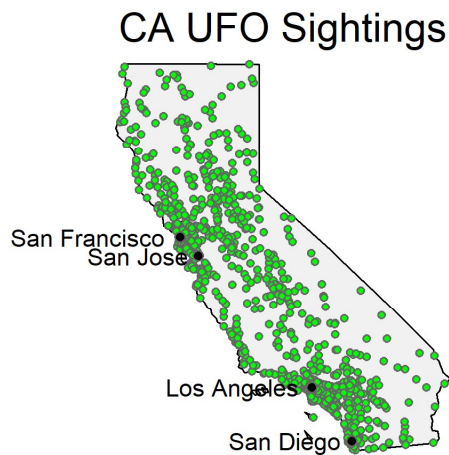
```
# set display params
par(mar=c(0,5,0,5))
# Make map displaying the California border and UFO sightings
ca_map <- tm_shape(cali) +
  tm_fill('NAME', palette='Greys', alpha=0.5) +
  tm_borders(col='black', lwd=1) +
  tm_shape(ufo_ca_sf) +
  tm_symbols(0.1, col='green') +
  tm_shape(cali_mcities) +
  tm_symbols(0.2, col='black') +
  tm_text("AREANAME", size=0.9, just=1.1) +
  tm_layout(main.title = "CA UFO Sightings", legend.show=F, main.title.position
='center', frame=F) +
  tmap_options(check.and.fix = TRUE)
ca_map
```



By adding another map alongside the UFO sightings, showing all public airports in California, it is possible to see correlation between sightings and airports.

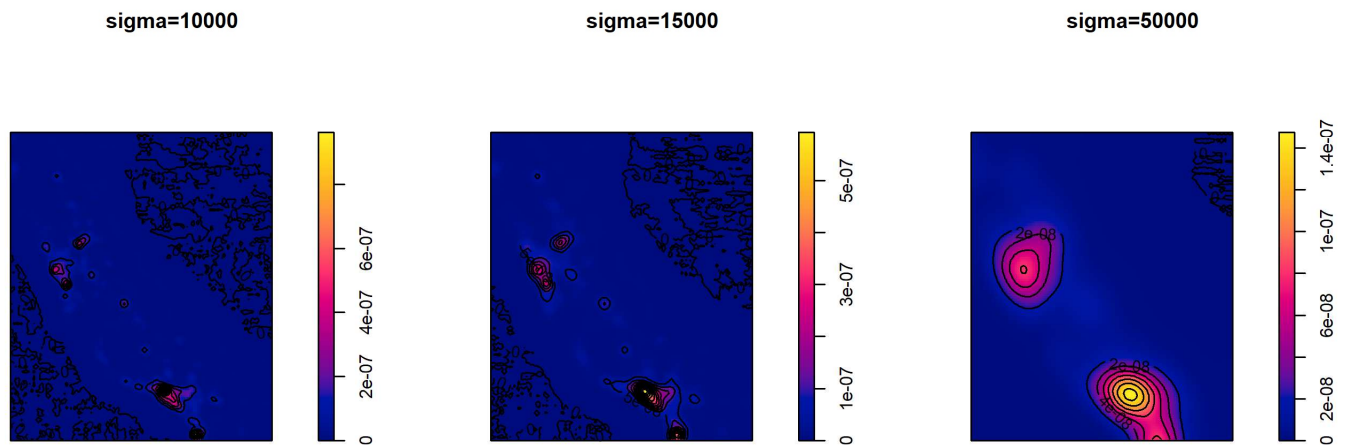
```
# Make maps displaying airports, military bases, and UFO sightings
```

```
ca_map2 <- tm_shape(cali) +  
  tm_fill('NAME', palette='Greys', alpha=0.5) +  
  tm_borders(col='black', lwd=1) +  
  tm_shape(bases_ca) +  
  tm_polygons(col='yellow') +  
  tm_shape(cali_airports) +  
  tm_symbols(0.1, col='red') +  
  tm_layout(main.title = "CA Airports and Bases", legend.show=F, main.title.position  
='center', frame=F)  
tmap_arrange(ca_map, ca_map2)
```



To view the KDE density plots, we simply use the same code from the *Methods* sections. So, I won't bother pasting it again here. For the sake of saving space, I will also not include all six sigma values. Here I am using 10,000, 15,000, and 50,000.

```
par(mfrow = c(1,3))  
d10K <- density(ufo_ca_ppp, sigma = 10000)  
plot(d10K, main="sigma=10000")  
contour(d10K, add = T)  
  
d15K <- density(ufo_ca_ppp, sigma = 15000)  
plot(d15K, main="sigma=15000")  
contour(d15K, add = T)  
  
d20K <- density(ufo_ca_ppp, sigma = 50000)  
plot(d20K, main="sigma=50000")  
contour(d20K, add = T)
```



For the ANN analysis, we first have to view the ANN index value.

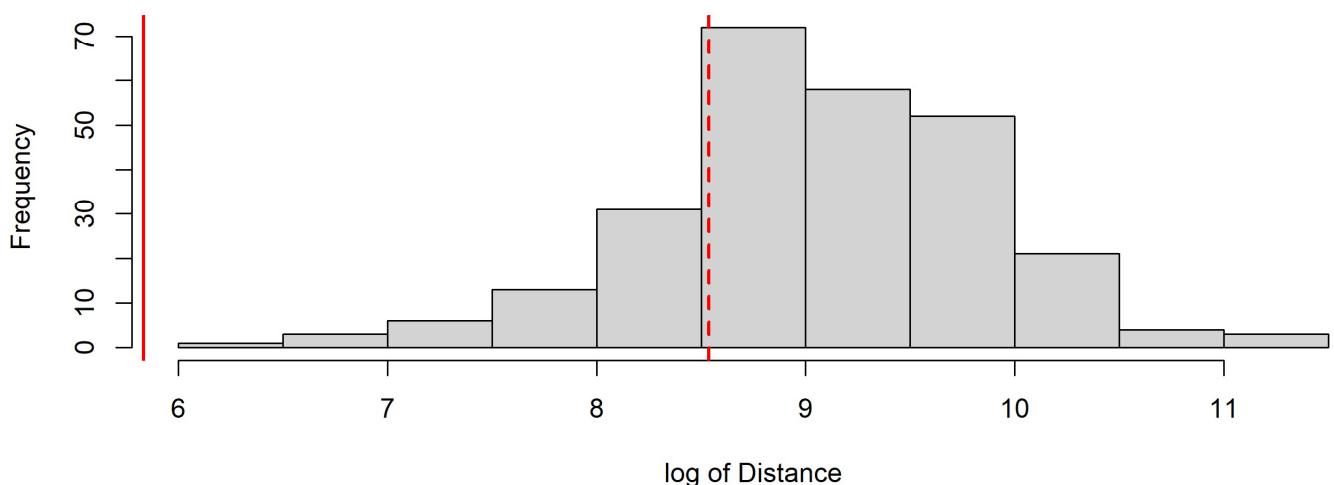
```
# View the ANN
ann_ufo

## [1] 0.06721282
```

Since this value is less than one, it indicates that the points are quite clustered and not randomly dispersed. We can see this when we plot it on a histogram as well. The dashed lined is the expected ANN index, which should be about random, while the solid line is the observed value, once again indicating clustering.

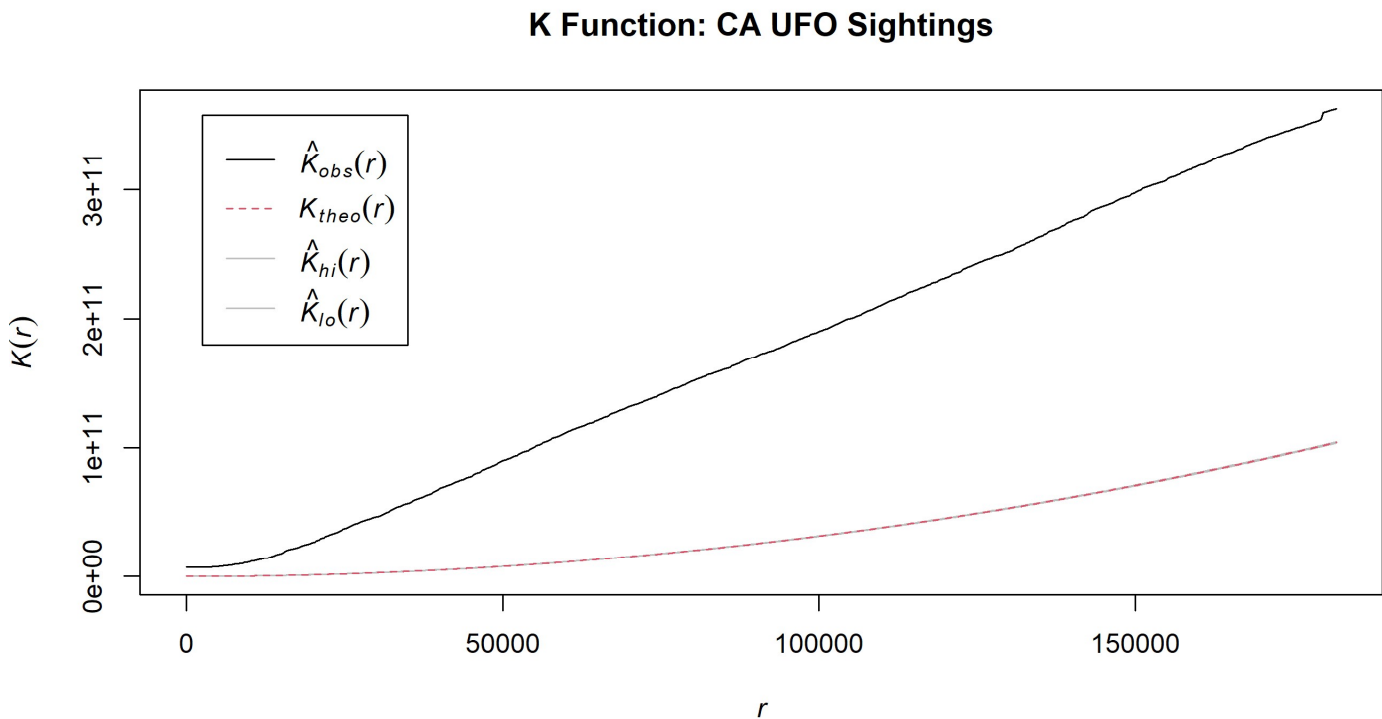
```
# Mean indicated with red line, expected mean indicated with dashed red line
hist(log(nnd_ufo), main = "Distances Between UFO Sightings\n ANN Index = 0.06721282",
xlab="log of Distance")
abline(v=log(mnnd_ufo), col='red', lwd=2)
abline(v=log(emnnd_ufo), col='red', lwd=2, lty='dashed')
```

Distances Between UFO Sightings
ANN Index = 0.06721282



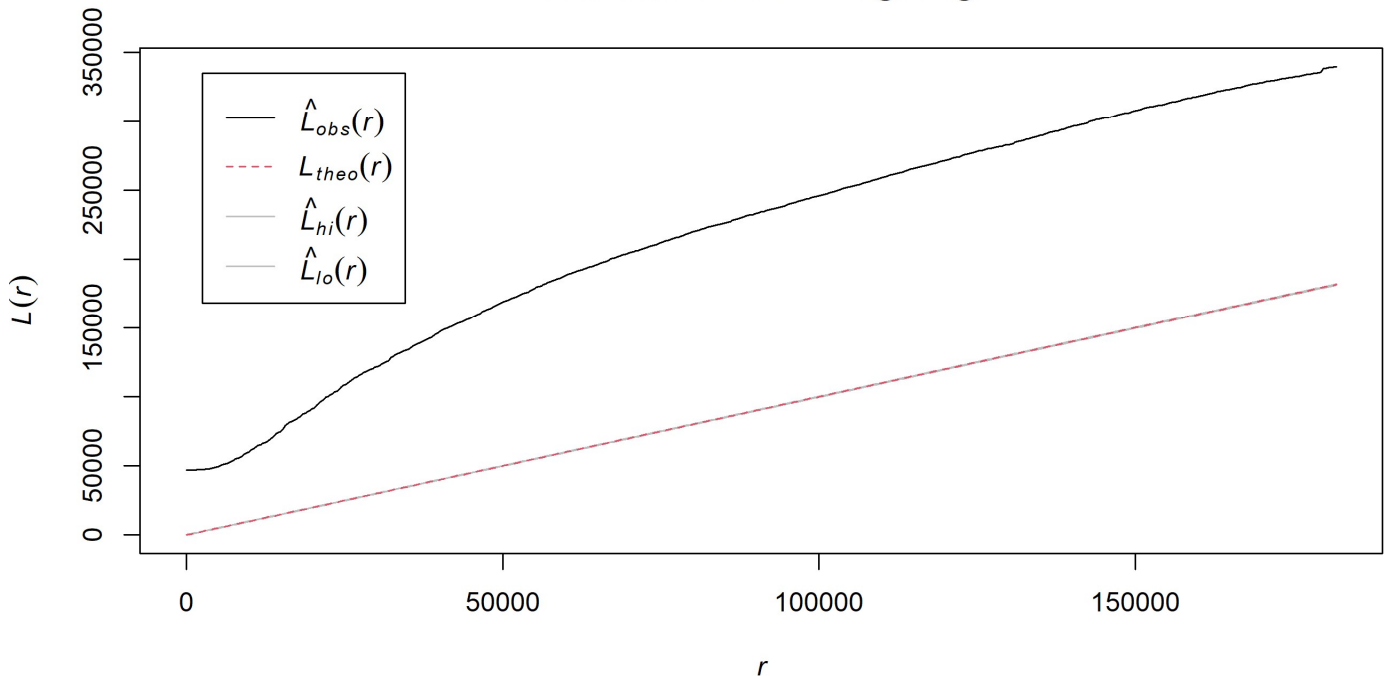
Moving on to the final point pattern analysis method - Ripley's K and L functions. We already created the K-function and L-function and simply need to use `plot()` to create a plot. In both plots it is obvious that the observed data, the solid line, is far above the expected, dashed, line. Once again indicating a non-random dispersion of points.

```
# plot the functions  
plot(k_ufo, main="K Function: CA UFO Sightings")
```



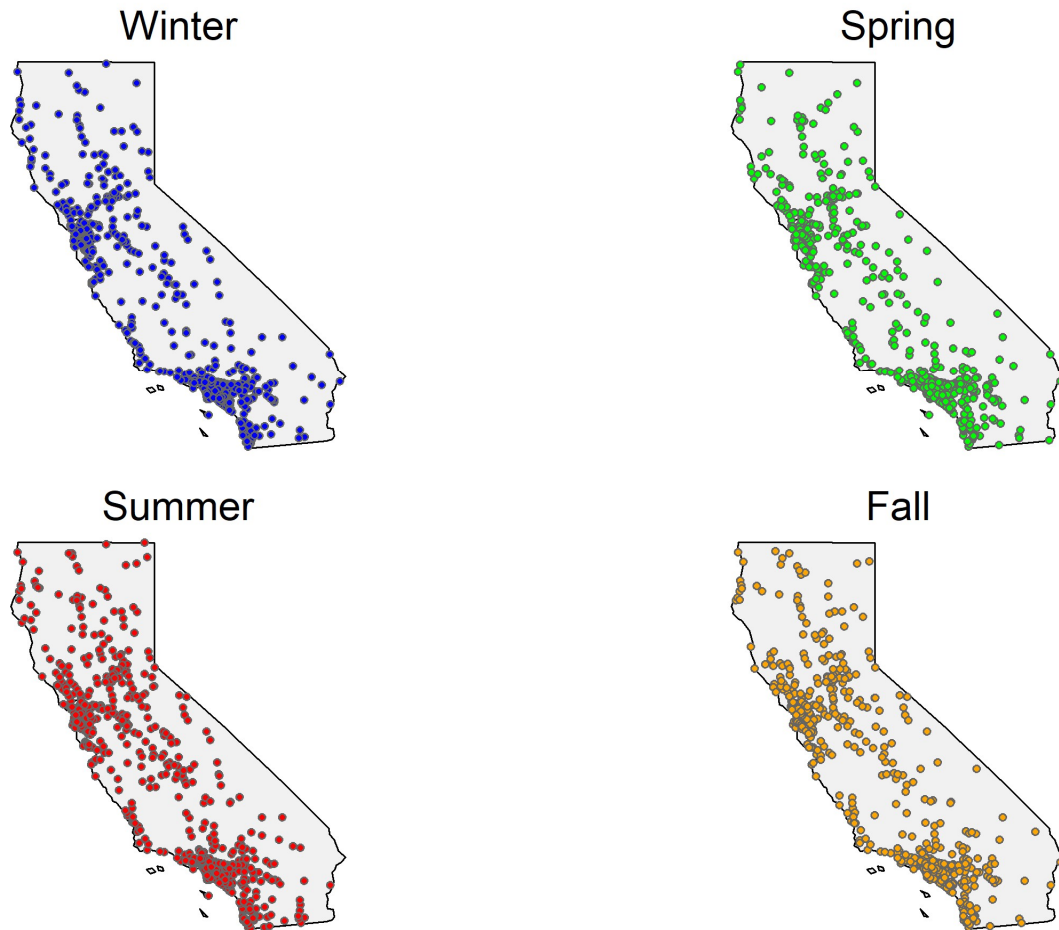
```
plot(l_ufo, main="L Function: CA UFO Sightings")
```

L Function: CA UFO Sightings



Finally, I plotted my 'seasons' variable. While I wasn't able to run any analyses on these variable in conjunction with others, I still think it is valuable to look at the information. From the plot below, it not possible to see any correlation between the season of occurrence and UFO sightings.

```
map_w <- tm_shape(cali) +
  tm_fill('NAME', palette='Greys', alpha=0.5) +
  tm_borders(col='black', lwd=1) +
  tm_shape(winter) +
  tm_symbols(0.1, col='blue') +
  tm_layout(main.title = "Winter", legend.show=F, main.title.position = 'center', frame=F)
map_sp <- tm_shape(cali) +
  tm_fill('NAME', palette='Greys', alpha=0.5) +
  tm_borders(col='black', lwd=1) +
  tm_shape(spring) +
  tm_symbols(0.1, col='green') +
  tm_layout(main.title = "Spring", legend.show=F, main.title.position = 'center', frame=F)
map_s <- tm_shape(cali) +
  tm_fill('NAME', palette='Greys', alpha=0.5) +
  tm_borders(col='black', lwd=1) +
  tm_shape(summer) +
  tm_symbols(0.1, col='red') +
  tm_layout(main.title = "Summer", legend.show=F, main.title.position = 'center', frame=F)
map_f <- tm_shape(cali) +
  tm_fill('NAME', palette='Greys', alpha=0.5) +
  tm_borders(col='black', lwd=1) +
  tm_shape(fall) +
  tm_symbols(0.1, col='orange') +
  tm_layout(main.title = "Fall", legend.show=F, main.title.position = 'center', frame=F)
tmap_arrange(map_w, map_sp, map_s, map_f, nrow=2, ncol=2)
```



Let's also look at the number of sightings per season. This is easily done with the `length()` function, which counts the number of observations in a specified variable.

```
length(winter$season)
## [1] 2028

length(spring$season)
## [1] 1913

length(summer$season)
## [1] 2639

length(fall$season)
## [1] 2332
```

So, returning to the project questions, what are the answers?

Is there a correlation between military bases/airports and UFO sightings? There is correlation between California public airports and California UFO sightings. However, this does not mean one equals the other. While it is most likely that aircraft activity lead to UFO sightings, it must be acknowledged that airports are where people live, the same people who make UFO observations. There simply are no people where there are no airports. Therefore, there will be no sightings.

Do seasons have an effect on the number of UFO sightings? No, what season a UFO sighting occurred in does not seem to have an effect. There appears to be an even distribution of UFO sightings in California across all four seasons, both spatially and statistically. While there are more sightings in summer and fall, I don't believe this to be indicative of any correlation.

Discussion

During my proposal, I was very motivated to do this project. I mean, who doesn't want to work with UFO sightings?? However, once I began my work, I began losing motivation and intrigue. When splitting the workload, Christine and I agreed that I would handle the seasons, spatial multivariate clustering, and ANN, while she would handle KDE and Ripley's K & L; little did I know the challenges I would encounter. In my opinion, Christine had the much easier set of tasks, allowing her to complete it without much struggle, while I spent hours in one sitting trying to fix one issue. Fortunately, The point pattern analyses worked almost flawlessly (from my understanding). By no means am I trying to put down Christine's hard work, she actually figured some things out, such as the ANN area calculation, which were extremely helpful, I am simply trying to paint a complete picture of this project.

Expanding on how work was distributed, I had expressed my apologies for not being able to get multivariate clustering working (even though I believe it was out of my control) and Christine suggested that I could write the *Methods* section of the report to compensate for my output. While I am happy to do so, as I would be writing it anyways, it was annoying to be asked because I had spent an enormous amount of time trying to get things to work. It's definitely not serious, but I thought it was interesting. So, all text in this report are written by me.

Regarding the substance of the project itself. I went into the project without serious expectations or hyper-scientific thought processes - I just wanted to take a surface-level look at UFO sightings data. I believe Christine and I achieved just that. While not everything worked, we were able to provide some analyses, along with bringing in other spatial data to assist with reviewing UFO sightings. If I were to do the project in the future, I would focus in on a much smaller area. I believe when working with nearly 9,000 points in an area as big as California, it can be difficult to run analyses and draw conclusions. Additionally, I think a case study-like project would be better for this kind of data. For example, I think looking at UFO sightings around Area 51 would be really fun and interesting.

Overall, I think the project was surprisingly successful. Even though a lot of the attempted analyses methods, from our labs and outside, did not work correctly due to the data itself, I believe the end result demonstrates myself and Christine's ability to adapt to a difficult situation. In the future, I would like to attempt a similar project that is more focused, like a case study. Additionally, while working with a teammate was beneficial, I personally believe I work better on my own as I don't have the stress of needing to work with another person along the way. However, I initially accepted Christine's offer for a partnership so I could get better at working with fellow GIS students. In the end, I believe the project was a success, despite the challenges myself and Christine faced along the way.

References

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